ML101
Homework 01: A 'gentle' Introduction to Machine Learning.

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1 Introduction

Machine Learning relies on a blend of statistical foundations, algorithmic design, and domain intuition. To kick off the semester, this assignment provides a structured framework to explore your first textbook chapter.

Through this task, you will engage with state-of-the-art AI tools to extract key takeaways, probe deeper into complex concepts through targeted questions, and visually organize your insights in a handwritten mind map. This approach is intended to strengthen both your technical curiosity and your ability to critically evaluate AI-generated knowledge.

2 Reading Material

We will select **one** of the following textbooks to cover the foundations and core principles of machine learning:

1. *Understanding Machine Learning: From Theory to Algorithms* by Shai Shalev-Shwartz and Shai Ben-David.
2. *An Introduction to Statistical Learning: with Applications in R* (or Python) by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor.
3. *Probabilistic Machine Learning: An Introduction* by Kevin P. Murphy.

3 Key Takeaways

We will use an AI tool to summarize Chapter 1 of our selected textbook. Feel free to request a translation of the content if you prefer reading in another language. Please keep a record of all prompts used to generate and refine your summary.

Additionally, ask the AI three (3) follow-up questions to explore specific concepts in greater depth. Be sure to document the exact prompts used for each question.

4 Individual Assignment

Based on the summary and the answers to your follow-up questions, create a mind map to illustrate your key findings and main concepts. The mind map must be **handwritten**.

5 Expected Deliverable

You will prepare a well-structured report documenting and discussing all prompts used in Section 3, along with a photo or scan of your handwritten mind map from Section 4.

Please submit your report in **PDF format** through the corresponding Brightspace™ assignment portal by **August 13, 2026**.

Happy Reading! 😎